Energy, Heat and Work

Energy units, work and the first law

Name _____ Class _____ Date _____

EQUATIONS AND CONSTANTS

$E_k = \frac{1}{2}mv^2$ · $E_p = mgh$ · $g = 9.81\ m/s^2$ · $w = -P(ext)\Delta V$ · $\Delta U = q + w$

1 cal = 4.184 J · 1 kcal = 4.184 kJ · 1 kWh = 3.6×10^6 J · 1 L·atm = 101.325 J

Sign convention: q and w are positive when energy enters the system. Show units in every step.

A. Ideas

1. (a) Define energy, heat and work. (b) Classify each as an open, closed or isolated system: a pan of boiling water with no lid; a sealed balloon; hot soup in a sealed, perfectly insulated flask.

2. A bath of warm water (40 °C) and a cup of boiling water (100 °C). Which has the higher temperature? Which contains more thermal energy? Explain.

B. Kinetic and potential energy

3. Calculate the kinetic energy of: (a) a 2.00 kg ball moving at 3.00 m/s (b) a nitrogen molecule (mass 4.65×10^{-26} kg) moving at 515 m/s.

4. How much potential energy does a 5.00 kg box gain when it is lifted 2.00 m?

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C. Energy units

5. Convert: (a) 520 kcal to kJ (b) 1200 kJ to kcal (c) 2.5 kWh to J

D. Work and the first law

6. A gas is compressed from 10.0 L to 4.0 L by a constant external pressure of 2.00 atm. Calculate the work in joules and explain its sign.

7. Calculate ΔU for each system: (a) loses 300. J of heat while 150. J of work is done on it (b) absorbs 500. J of heat and does 200. J of work (c) a gas in a rigid, sealed container absorbs 75 J of heat.

E. Challenge

8. A gas in a cylinder with a piston absorbs 2.50 kJ of heat and expands from 1.00 L to 6.00 L against a constant external pressure of 1.00 atm. Calculate (a) the work done, in J (b) ΔU, in kJ. (c) Did the internal energy of the gas increase or decrease?

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